

CBSE CLASS X
Mathematics Standard (041)

QUESTION PAPER

From previous CBSE Board Exam questions

Code: AA7248**Questions: 15****Maximum Marks: 31****Generated: 2026-06-15 13:05****SELECTIONS USED**

Source	Previous-year board
Subject	Mathematics
Lessons	Proofs in Mathematics
Year	2022-2026
Questions selected	15

Q1. § 1.3 Revisiting Irrational Numbers**[2]**

Prove that $5 - 2\sqrt{3}$ is an irrational number. It is given that $\sqrt{3}$ is an irrational number.

Previously asked in: 2024 30/1/1 Q25(A)

◆ **Real Numbers** 1.3 Revisiting Irrational Numbers A1.7 Proof by Contradiction

Q2. § 1.3 Revisiting Irrational Numbers**[2]**

Prove that $2 - 5\sqrt{3}$ is an irrational number given that $\sqrt{3}$ is irrational.

Previously asked in: 2026 30/5/1 Q21

◆ **Real Numbers** 1.3 Revisiting Irrational Numbers A1.7 Proof by Contradiction

Q3. § 1.3 Revisiting Irrational Numbers**[3]**

Prove that $\sqrt{2}$ is an irrational number.

Previously asked in: 2025 30/4/1 Q26(A); 2023 30/6/1 Q27(B) – 2×

◆ **Real Numbers** 1.3 Revisiting Irrational Numbers A1.7 Proof by Contradiction

Q4. § 1.3 Revisiting Irrational Numbers § 1.4 Summary**[2]**

Prove that $2 + \sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is an irrational number.

Previously asked in: 2023 30/2/1 Q21

◆ **Real Numbers** 1.3 Revisiting Irrational Numbers A1.7 Proof by Contradiction

Q5. § 1.3 Revisiting Irrational Numbers § 1.4 Summary**[3]**

Prove that $\left(\sqrt{2} + \frac{\sqrt{3}}{2}\right)$ is an irrational number, given that $\sqrt{6}$ is an irrational number.

Previously asked in: 2024 30/5/1 Q26(b) (OR-2)

◆ **Real Numbers** 1.3 Revisiting Irrational Numbers A1.7 Proof by Contradiction

Q6. § 1.3 Revisiting Irrational Numbers**[3]**

Prove that $\frac{2 - \sqrt{3}}{5}$ is an irrational number, given that $\sqrt{3}$ is an irrational number.

Previously asked in: 2024 30/4/1 Q30

◆ **Real Numbers** 1.3 Revisiting Irrational Numbers A1.7 Proof by Contradiction

Q7. § 1.3 Revisiting Irrational Numbers § 1.4 Summary**[3]**

Prove that $4 + 3\sqrt{2}$ is an irrational number given that $\sqrt{2}$ is an irrational number.

Previously asked in: 2025 30/3/1 Q27

♦ **Real Numbers** 1.3 Revisiting Irrational Numbers A1.7 Proof by Contradiction

Q8. § 1.3 Revisiting Irrational Numbers § 1.4 Summary**[3]**

Prove that $\sqrt{3}$ is an irrational number.

Previously asked in: 2026 30/1/1 Q26; 2025 30/2/1 Q30; 2025 30/5/1 Q26(a); 2024 30/5/1 Q26(a) (OR-1); 2023 30/5/1 Q26 (OR-1) – 5×

♦ **Real Numbers** 1.3 Revisiting Irrational Numbers A1.7 Proof by Contradiction

Q9. § 1.3 Revisiting Irrational Numbers**[3]**

Prove that $\sqrt{5}$ is an irrational number.

Previously asked in: 2026 30/2/1 Q26; 2026 30/3/1 Q28; 2025 30/1/1 Q29; 2025 30/6/1 Q26(a); 2024 30/2/1 Q27; 2023 30/4/1 Q27; 2023 30/1/1 Q27 – 7×

♦ **Real Numbers** 1.3 Revisiting Irrational Numbers A1.7 Proof by Contradiction

Q10. § 10.1 Introduction**[1]**

Questions number 19 and 20 are Assertion and Reason based questions carrying 1 mark each. Select the correct answer from codes (a)–(d).

Assertion (A) : If the points $A(4, 3)$ and $B(x, 5)$ lie on a circle with centre $O(2, 3)$, then the value of x is 2.

Reason (R) : Centre of a circle is the mid-point of each chord of the circle.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
- (c) Assertion (A) is true, but Reason (R) is false.
- (d) Assertion (A) is false, but Reason (R) is true.

Previously asked in: 2023 30/5/1 Q19

♦ **Circles** A1.2 Mathematical Statements Revisited Points on a Circle — Equal Radii Property

Q11. § A1.8 Summary**[1]**

Which of the following statements is false ?

- A Two right triangles are always similar.
- B Two squares are always similar.
- C Two equilateral triangles are always similar.
- D Two circles are always similar.

Previously asked in: 2025 30/5/1 Q8; 2025 30/5/1 Q10 – 2×

♦ **Proofs in Mathematics** A1.2 Mathematical Statements Revisited

Q12. § A1.8 Summary**[1]**

Which of the following statements is true ?

- (a) $\sin 20^\circ > \sin 70^\circ$
- (b) $\sin 20^\circ > \cos 20^\circ$
- (c) $\cos 20^\circ > \cos 70^\circ$
- (d) $\tan 20^\circ > \tan 70^\circ$

Previously asked in: 2025 30/4/1 Q9

♦ **Proofs in Mathematics** A1.2 Mathematical Statements Revisited

Q13.**[1]**

The natural number 2 is :

- (a) a prime number
- (b) a composite number
- (c) prime as well as composite
- (d) neither prime nor composite

Previously asked in: 2026 30/1/1 Q2

♦ Proofs in Mathematics

A1.2 Mathematical Statements Revisited

Q14.

§ A1.2 Mathematical Statements Revisited

§ A1.5 Negation of a Statement

§ A1.8 Summary

[1]

Which of the following statements is incorrect ?

- A Two congruent figures are always similar.
- B A square and a rhombus of the same area are always similar.
- C Two equilateral triangles are always similar.
- D Two similar triangles need not be congruent.

Previously asked in: 2025 30/2/1 Q9

♦ Proofs in Mathematics

A1.2 Mathematical Statements Revisited

Q15.**[2]**

Show that the number $5 \times 11 \times 17 + 3 \times 11$ is a composite number.

Previously asked in: 2024 30/1/1 Q25(B)

♦ Proofs in Mathematics

Composite Numbers and Factorization Proofs

CBSE CLASS X

Mathematics Standard (041)

ANSWER KEY

From previous CBSE Board Exam questions

Code: AA7248

Questions: 15

Maximum Marks: 31

Generated: 2026-06-15 13:05

Q1. § 1.3 Revisiting Irrational Numbers

[2]

Prove that $5 - 2\sqrt{3}$ is an irrational number. It is given that $\sqrt{3}$ is an irrational number.

Previously asked in: 2024 30/1/1 Q25(A)

◆ Real Numbers

1.3 Revisiting Irrational Numbers

A1.7 Proof by Contradiction

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Model Answer

Proof (by contradiction):

Assume $5 - 2\sqrt{3}$ is rational. Then we can write:

$$5 - 2\sqrt{3} = \frac{a}{b}, \quad \text{where } a, b \text{ are integers, } b \neq 0$$

Rearranging:

$$2\sqrt{3} = 5 - \frac{a}{b} = \frac{5b - a}{b}$$
$$\sqrt{3} = \frac{5b - a}{2b}$$

Since a and b are integers, $\frac{5b - a}{2b}$ is rational, which means $\sqrt{3}$ is rational.But this **contradicts** the given fact that $\sqrt{3}$ is irrational.Therefore, our assumption is wrong, and $5 - 2\sqrt{3}$ is **irrational**. ■

Source: Real Numbers, Section 1.3 – Revisiting Irrational Numbers

Explanation

- Use **proof by contradiction**: assume the number is rational, express it as $\frac{a}{b}$, and rearrange to isolate $\sqrt{3}$.
- The key step is showing this forces $\sqrt{3}$ to be rational – contradicting the given condition.
- Examiners award marks for: (1) correct assumption, (2) correct algebraic rearrangement to isolate $\sqrt{3}$, (3) stating the contradiction clearly.
- Don't forget to state the conclusion explicitly.

Q2. § 1.3 Revisiting Irrational Numbers**[2]**

Prove that $2 - 5\sqrt{3}$ is an irrational number given that $\sqrt{3}$ is irrational.

Previously asked in: 2026 30/5/1 Q21

♦ Real Numbers

1.3 Revisiting Irrational Numbers

A1.7 Proof by Contradiction

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Model Answer

Assume, to the contrary, that $2 - 5\sqrt{3}$ is rational.

Then there exist coprime integers a and b ($b \neq 0$) such that:

$$\begin{aligned}2 - 5\sqrt{3} &= \frac{a}{b} \\ \Rightarrow 5\sqrt{3} &= 2 - \frac{a}{b} = \frac{2b - a}{b} \\ \Rightarrow \sqrt{3} &= \frac{2b - a}{5b}\end{aligned}$$

Since a and b are integers, $\frac{2b - a}{5b}$ is rational, which means $\sqrt{3}$ is rational.

But this **contradicts** the given fact that $\sqrt{3}$ is irrational.

Therefore, our assumption is wrong and $2 - 5\sqrt{3}$ is **irrational**.

Source: Chapter 1, Section 1.3 — Revisiting Irrational Numbers

Explanation

- This is a **proof by contradiction** question — always start by assuming the opposite of what you want to prove.
- The key step is isolating $\sqrt{3}$ on one side to show it would have to be rational — contradicting the given condition.
- Examiners award marks for: (1) correct assumption, (2) algebraic rearrangement to isolate $\sqrt{3}$, (3) stating the contradiction clearly. Don't skip the contradiction statement.

Q3. § 1.3 Revisiting Irrational Numbers**[3]**Prove that $\sqrt{2}$ is an irrational number.

Previously asked in: 2025 30/4/1 Q26(A); 2023 30/6/1 Q27(B) – 2×

◆ Real Numbers

1.3 Revisiting Irrational Numbers

A1.7 Proof by Contradiction

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Model Answer**Proof:** Assume, to the contrary, that $\sqrt{2}$ is rational.Then we can find integers a and b ($b \neq 0$) such that $\sqrt{2} = \frac{a}{b}$, where a and b are coprime (no common factor other than 1).Squaring both sides: $2b^2 = a^2$ So 2 divides a^2 . By Theorem 1.2, 2 divides a .Let $a = 2c$ for some integer c . Substituting:

$$2b^2 = 4c^2 \implies b^2 = 2c^2$$

So 2 divides b^2 , which means 2 divides b .Thus 2 is a common factor of both a and b , which **contradicts** the fact that a and b are coprime.This contradiction arose because of our incorrect assumption. Hence, $\sqrt{2}$ is irrational.

Source: Chapter 1, Section 1.3 – Revisiting Irrational Numbers

Explanation

- This is a **proof by contradiction** — always start by assuming the opposite of what you want to prove.
- Key step: use **Theorem 1.2** (if a prime p divides a^2 , then p divides a) — examiners expect you to cite this.
- The contradiction must be clearly stated: a and b were assumed coprime, but both are divisible by 2.
- Write "Hence $\sqrt{2}$ is irrational" as the concluding line — never omit the conclusion.

Q4. § 1.3 Revisiting Irrational Numbers § 1.4 Summary

[2]

Prove that $2 + \sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is an irrational number.

Previously asked in: 2023 30/2/1 Q21

◆ Real Numbers

1.3 Revisiting Irrational Numbers

A1.7 Proof by Contradiction

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Model Answer**Proof (by contradiction):**Assume $2 + \sqrt{3}$ is rational. Then we can write:

$$2 + \sqrt{3} = \frac{a}{b}, \quad \text{where } a, b \text{ are integers and } b \neq 0$$

Rearranging:

$$\sqrt{3} = \frac{a}{b} - 2 = \frac{a - 2b}{b}$$

Since a and b are integers, $\frac{a - 2b}{b}$ is rational, which means $\sqrt{3}$ is rational.But this **contradicts** the given fact that $\sqrt{3}$ is irrational.Therefore, our assumption is wrong, and $2 + \sqrt{3}$ is **irrational**. ■

Source: Chapter 1, Section 1.3 – Revisiting Irrational Numbers

Explanation

- This is a standard "proof by contradiction" question. Always start by **assuming the opposite** of what you want to prove.
- The key step is **isolating** $\sqrt{3}$ on one side — showing it equals a rational expression, which contradicts the given condition.
- Examiners award marks for: (1) correct assumption, (2) correct rearrangement isolating $\sqrt{3}$, (3) stating the contradiction clearly, (4) conclusion. Don't skip the conclusion line.

Q5. § 1.3 Revisiting Irrational Numbers § 1.4 Summary

[3]

Prove that $\left(\sqrt{2} + \frac{\sqrt{3}}{2}\right)$ is an irrational number, given that $\sqrt{6}$ is an irrational number.

Previously asked in: 2024 30/5/1 Q26(b) (OR-2)

◆ Real Numbers 1.3 Revisiting Irrational Numbers A1.7 Proof by Contradiction

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Model Answer

Assume, to the contrary, that $\left(\sqrt{2} + \frac{\sqrt{3}}{2}\right)$ is rational.

Then there exist coprime integers a and b ($b \neq 0$) such that:

$$\sqrt{2} + \frac{\sqrt{3}}{2} = \frac{a}{b}$$

Rearranging:

$$\frac{\sqrt{3}}{2} = \frac{a}{b} - \sqrt{2}$$

$$\sqrt{3} = 2\left(\frac{a}{b} - \sqrt{2}\right) = \frac{2a}{b} - 2\sqrt{2}$$

$$2\sqrt{2} = \frac{2a}{b} - \sqrt{3}$$

$$\sqrt{6} = \frac{1}{2}\left(\frac{2a}{b} - \sqrt{3}\right) \cdot \sqrt{2}$$

A simpler path: squaring both sides of $\sqrt{2} + \frac{\sqrt{3}}{2} = \frac{a}{b}$:

$$2 + \sqrt{6} + \frac{3}{4} = \frac{a^2}{b^2}$$

$$\sqrt{6} = \frac{a^2}{b^2} - \frac{11}{4} = \frac{4a^2 - 11b^2}{4b^2}$$

Since a, b are integers, the RHS $\frac{4a^2 - 11b^2}{4b^2}$ is **rational**.

This means $\sqrt{6}$ is rational — **a contradiction**, since $\sqrt{6}$ is given to be irrational.

Hence, our assumption is wrong, and $\left(\sqrt{2} + \frac{\sqrt{3}}{2}\right)$ is **irrational**. ■

Source: Chapter 1, Section 1.3 – Revisiting Irrational Numbers

Explanation

- The standard method is **proof by contradiction**: assume the expression is rational, then algebraically isolate a known irrational ($\sqrt{6}$ here) and show it would have to be rational — contradiction.
- The key step is squaring: $\left(\sqrt{2} + \frac{\sqrt{3}}{2}\right)^2 = 2 + \sqrt{6} + \frac{3}{4} = \frac{11}{4} + \sqrt{6}$, which lets you isolate $\sqrt{6}$.

- Always explicitly state "this contradicts the fact that $\sqrt{6}$ is irrational" — examiners award a mark specifically for this conclusion.
- The given condition " $\sqrt{6}$ is irrational" **must** be used; never leave it unused.

Q6. § 1.3 Revisiting Irrational Numbers

[3]

Prove that $\frac{2 - \sqrt{3}}{5}$ is an irrational number, given that $\sqrt{3}$ is an irrational number.

Previously asked in: 2024 30/4/1 Q30

♦ Real Numbers

1.3 Revisiting Irrational Numbers

A1.7 Proof by Contradiction

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Model Answer**Proof by contradiction.**

Assume, to the contrary, that $\frac{2 - \sqrt{3}}{5}$ is rational.

Then we can find coprime integers a and b ($b \neq 0$) such that:

$$\frac{2 - \sqrt{3}}{5} = \frac{a}{b}$$

Rearranging:

$$2 - \sqrt{3} = \frac{5a}{b}$$

$$\sqrt{3} = 2 - \frac{5a}{b} = \frac{2b - 5a}{b}$$

Since a and b are integers, $\frac{2b - 5a}{b}$ is rational, which means $\sqrt{3}$ is rational.

But this **contradicts** the given fact that $\sqrt{3}$ is irrational.

This contradiction arose because of our incorrect assumption.

Therefore, $\frac{2 - \sqrt{3}}{5}$ is irrational. ■

Source: Chapter 1, Section 1.3 Revisiting Irrational Numbers

Explanation

- The key technique is **proof by contradiction**: assume the number is rational, then algebraically isolate $\sqrt{3}$ and show it would have to be rational — contradicting the given condition.
- Always state that a and b are integers to justify that the rearranged expression is rational.
- Examiners award marks for: (1) correct assumption, (2) correct algebraic rearrangement to isolate $\sqrt{3}$, (3) citing the contradiction with the given fact, and (4) the conclusion. Don't skip any step.

Q7. § 1.3 Revisiting Irrational Numbers § 1.4 Summary

[3]

Prove that $4 + 3\sqrt{2}$ is an irrational number given that $\sqrt{2}$ is an irrational number.

Previously asked in: 2025 30/3/1 Q27

◆ Real Numbers 1.3 Revisiting Irrational Numbers A1.7 Proof by Contradiction

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Model Answer**Proof (by contradiction):**Assume, to the contrary, that $4 + 3\sqrt{2}$ is rational.Then we can find coprime integers a and b ($b \neq 0$) such that:

$$4 + 3\sqrt{2} = \frac{a}{b}$$

Rearranging:

$$3\sqrt{2} = \frac{a}{b} - 4 = \frac{a - 4b}{b}$$

$$\sqrt{2} = \frac{a - 4b}{3b}$$

Since a and b are integers, $\frac{a - 4b}{3b}$ is rational, which means $\sqrt{2}$ is rational.But this **contradicts** the given fact that $\sqrt{2}$ is irrational.

This contradiction arose because of our incorrect assumption.

 $\therefore 4 + 3\sqrt{2}$ is irrational. ■

Source: Chapter 1, Section 1.3 – Revisiting Irrational Numbers

Explanation

- Use **proof by contradiction**: assume the number is rational, express it as $\frac{a}{b}$ with a, b coprime.
- Isolate $\sqrt{2}$ algebraically — it ends up equal to a rational expression, contradicting the given condition.
- Always state clearly *what* the contradiction is and *why* it arose.
- Examiners award marks for: correct assumption (1), correct algebraic rearrangement to isolate $\sqrt{2}$ (1), stating the contradiction and conclusion (1).

Q8. § 1.3 Revisiting Irrational Numbers § 1.4 Summary

[3]

Prove that $\sqrt{3}$ is an irrational number.

Previously asked in: 2026 30/1/1 Q26; 2025 30/2/1 Q30; 2025 30/5/1 Q26(a); 2024 30/5/1 Q26(a) (OR-1); 2023 30/5/1 Q26 (OR-1) — 5×

♦ Real Numbers 1.3 Revisiting Irrational Numbers A1.7 Proof by Contradiction

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Model Answer**Proof:** Assume, to the contrary, that $\sqrt{3}$ is rational.Then we can find integers a and b ($b \neq 0$) such that $\sqrt{3} = \frac{a}{b}$, where a and b are coprime (no common factor other than 1).So, $b\sqrt{3} = a$.Squaring both sides: $3b^2 = a^2$ Therefore, a^2 is divisible by 3, and by Theorem 1.2, a is also divisible by 3.So, let $a = 3c$ for some integer c .Substituting: $3b^2 = 9c^2 \Rightarrow b^2 = 3c^2$ This means b^2 is divisible by 3, so b is also divisible by 3.Therefore, a and b have at least 3 as a common factor. But this **contradicts** the fact that a and b are coprime.This contradiction arose because of our incorrect assumption. Hence, $\sqrt{3}$ is irrational. ■

Source: Chapter 1, Section 1.3 – Revisiting Irrational Numbers

Explanation

- This is a **proof by contradiction** — always start by assuming the opposite of what you want to prove.
- The key tool used is **Theorem 1.2**: if a prime p divides a^2 , then p divides a .
- The critical step students miss: after substituting $a = 3c$, show that **3 also divides b** — this creates the contradiction with "coprime."
- Write the conclusion clearly: state the contradiction and then state that $\sqrt{3}$ is irrational.
- Examiners award marks for: assumption (½), algebraic steps (1), applying Theorem 1.2 twice (½), contradiction + conclusion (1).

Q9. § 1.3 Revisiting Irrational Numbers

[3]

Prove that $\sqrt{5}$ is an irrational number.

Previously asked in: 2026 30/2/1 Q26; 2026 30/3/1 Q28; 2025 30/1/1 Q29; 2025 30/6/1 Q26(a); 2024 30/2/1 Q27; 2023 30/4/1 Q27; 2023 30/1/1 Q27 — 7×

♦ Real Numbers

1.3 Revisiting Irrational Numbers

A1.7 Proof by Contradiction

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Model Answer

Proof: Assume, to the contrary, that $\sqrt{5}$ is rational.Then we can find integers a and b ($b \neq 0$) such that $\sqrt{5} = \frac{a}{b}$, where a and b are coprime.Squaring both sides: $5b^2 = a^2$ So 5 divides a^2 . By Theorem 1.2, 5 divides a .Write $a = 5c$ for some integer c . Substituting:

$$5b^2 = 25c^2 \Rightarrow b^2 = 5c^2$$

So 5 divides b^2 , and therefore 5 divides b .Thus 5 is a common factor of a and b , **contradicting** the fact that a and b are coprime.This contradiction arose because we assumed $\sqrt{5}$ is rational. Hence, $\sqrt{5}$ is irrational. ■

Source: Chapter 1, Section 1.3 – Revisiting Irrational Numbers

Explanation

- This is a standard **proof by contradiction**. Always start by assuming the opposite of what you want to prove.
- The key tool is **Theorem 1.2**: if a prime p divides a^2 , then p divides a . Examiners expect you to cite or use this explicitly.
- The contradiction must target the **coprime condition** — that is where the marks are.
- Write the substitution step ($a = 5c$) clearly; missing it loses a mark.

Q10. § 10.1 Introduction

[1]

Questions number 19 and 20 are Assertion and Reason based questions carrying 1 mark each. Select the correct answer from codes (a)–(d).

Assertion (A) : If the points $A(4, 3)$ and $B(x, 5)$ lie on a circle with centre $O(2, 3)$, then the value of x is 2.

Reason (R) : Centre of a circle is the mid-point of each chord of the circle.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
- (c) Assertion (A) is true, but Reason (R) is false.
- (d) Assertion (A) is false, but Reason (R) is true.

Previously asked in: 2023 30/5/1 Q19

◆ Circles

A1.2 Mathematical Statements Revisited

Points on a Circle — Equal Radii Property

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Model Answer

(c) Assertion (A) is true, but Reason (R) is false.

Since $OA = OB$ (radii), $\sqrt{(4-2)^2 + (3-3)^2} = \sqrt{(x-2)^2 + (5-3)^2}$ gives $2 = \sqrt{(x-2)^2 + 4}$, so $x = 2$. ✓

The centre is NOT the mid-point of every chord; it bisects a chord only when the line from centre is perpendicular to it.

Explanation

- **Assertion check:** $OA = \sqrt{4+0} = 2$; $OB = \sqrt{(x-2)^2 + 4} = 2 \Rightarrow (x-2)^2 = 0 \Rightarrow x = 2$. True.
- **Reason check:** The centre is the mid-point of a chord **only if** it is the perpendicular bisector of that chord — it is not the mid-point of *every* chord. The statement as given is false.
- So the answer is **(c)**: A is true, R is false.

Q11. § A1.8 Summary

[1]

Which of the following statements is false ?

- A Two right triangles are always similar.
- B Two squares are always similar.
- C Two equilateral triangles are always similar.
- D Two circles are always similar.

Previously asked in: 2025 30/5/1 Q8; 2025 30/5/1 Q10 — 2×

◆ Proofs in Mathematics

A1.2 Mathematical Statements Revisited

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Model Answer

Option A — Two right triangles are always similar is false.

Right triangles have one angle = 90° , but the other angles can differ, so they need not be similar.

Explanation

- Two figures are similar if all corresponding angles are equal and sides are proportional.
- Right triangles share one 90° angle, but the remaining two angles can vary (e.g., 30° - 60° - 90° vs. 45° - 45° - 90°), so similarity is not guaranteed.
- Squares, equilateral triangles, and circles are always similar because their shapes are fixed regardless of size.

Q12. § A1.8 Summary

[1]

Which of the following statements is true ?

- (a) $\sin 20^\circ > \sin 70^\circ$
- (b) $\sin 20^\circ > \cos 20^\circ$
- (c) $\cos 20^\circ > \cos 70^\circ$
- (d) $\tan 20^\circ > \tan 70^\circ$

Previously asked in: 2025 30/4/1 Q9

♦ Proofs in Mathematics

A1.2 Mathematical Statements Revisited

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Model Answer

(c) $\cos 20^\circ > \cos 70^\circ$

As the angle increases from 0° to 90° , cosine decreases. So $\cos 20^\circ > \cos 70^\circ$. (sin and tan increase, making options a, b, d false.)

Explanation

Key rule: **sin and tan are increasing functions** on $(0^\circ, 90^\circ)$, while **cos is a decreasing function**. So a smaller angle gives a larger cosine value — hence $\cos 20^\circ > \cos 70^\circ$ is the only true statement. Eliminate the others: $\sin 20^\circ < \sin 70^\circ$ (sin increases), $\cos 20^\circ > \sin 20^\circ$ only when angle $< 45^\circ$ but that's not what option (b) says — $\sin 20^\circ < \cos 20^\circ$, and $\tan 20^\circ < \tan 70^\circ$ (tan increases). Examiners expect you to recall increasing/decreasing behaviour of trig ratios.

Q13.

[1]

The natural number 2 is :

- (a) a prime number
- (b) a composite number
- (c) prime as well as composite
- (d) neither prime nor composite

Previously asked in: 2026 30/1/1 Q2

♦ Proofs in Mathematics

A1.2 Mathematical Statements Revisited

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Model Answer

(a) a prime number

2 is a prime number because it has exactly two distinct factors: 1 and itself, and it is listed among the collection of prime numbers in the Fundamental Theorem of Arithmetic.

Source: Chapter 1, Section 1.2

Explanation

2 is the smallest and the only even prime number. A prime number has exactly two factors — 1 and itself. 2 satisfies this condition. It is neither composite (which requires more than two factors) nor both. Examiners expect the correct option with a brief justification for MCQs.

Q14. § A1.2 Mathematical Statements Revisited § A1.5 Negation of a Statement § A1.8 Summary

[1]

Which of the following statements is incorrect ?

- A Two congruent figures are always similar.
- B A square and a rhombus of the same area are always similar.
- C Two equilateral triangles are always similar.
- D Two similar triangles need not be congruent.

Previously asked in: 2025 30/2/1 Q9

♦ Proofs in Mathematics A1.2 Mathematical Statements Revisited

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Model Answer

Answer: (B)

A square and a rhombus of the same area are **not** always similar, as their angles differ (a square has all 90° angles, a rhombus need not).

Explanation

Similarity requires both equal corresponding angles AND proportional sides. A rhombus has equal sides but its angles can vary, so it is not always similar to a square. The other three statements are correct: congruent figures are always similar (ratio 1:1); two equilateral triangles always have 60° - 60° - 60° ; similar triangles need not be congruent. Examiners expect you to identify the false statement and briefly justify.

Q15.

[2]

Show that the number $5 \times 11 \times 17 + 3 \times 11$ is a composite number.

Previously asked in: 2024 30/1/1 Q25(B)

♦ Proofs in Mathematics Composite Numbers and Factorization Proofs

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Model Answer

$$\begin{aligned}
 &5 \times 11 \times 17 + 3 \times 11 \\
 &= 11(5 \times 17 + 3) \\
 &= 11(85 + 3) \\
 &= 11 \times 88 \\
 &= 11 \times 8 \times 11 \\
 &= 8 \times 11^2
 \end{aligned}$$

Since the number can be expressed as a product of factors (other than 1 and itself), it is a **composite number**.

Source: Chapter 1, The Fundamental Theorem of Arithmetic

Explanation

- The key step is **taking 11 as a common factor**. This immediately shows the number is divisible by 11 (and also by 8), so it has factors other than 1 and itself — the definition of a composite number.
- Examiners expect you to show the factorisation step-by-step and explicitly state the conclusion. Do not just calculate the final value (968); show *why* it is composite by exhibiting its factors.
- This is similar to Exercise 1.1, Q.6 from Chapter 1.